

INNOVATIVE WAYS TO INCREASE PRODUCTION EFFICIENCY IN THE UZBEK FOOD INDUSTRY IN THE FACE OF GLOBAL CHALLENGES

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ABSTRACT

This study examines ways to improve production efficiency in Uzbekistan's food industry amid global challenges such as resource scarcity, climate change, and food security. Data from the State Statistics Committee, FAO, World Bank, and OECD (2020–2024) show production growth, but efficiency remains below international standards due to outdated technologies, low labor productivity, and inefficient resource use. Comparative analysis points to Turkey's halal certification system, which strengthened exports and consumer trust, and the Netherlands' circular food model, which combined economic and environmental efficiency through recycling. For Uzbekistan, priority measures include technological modernization, energy and resource efficiency, adoption of international certification, human capital development, and stronger state support. These steps can enhance food security, competitiveness, and sustainable growth.

INTRODUCTION

In recent years, global challenges such as climate change, resource scarcity, food security, and energy efficiency have had a direct impact on production processes. According to UN forecasts, the world food system will need to feed 9.7 billion people by 2050¹. This will require increased production efficiency, rational use of resources, and widespread adoption of innovative technologies. These issues are also relevant for Uzbekistan. The country's population is growing by an average of 700,000 people per year², which significantly increases domestic demand for food products. At the same time, there is a need to increase export potential, reduce import dependence, and optimize production costs. The food industry, in particular, the production of meat and sausage products, is one of the strategic segments of the economy and occupies an important place in the consumer basket of the population. Developing

innovative ways to increase efficiency in this area will serve to strengthen the country's economic security, population health, and competitiveness in the global market³. Therefore, this article aims to analyze innovative ways to increase the efficiency of production in the Uzbek food industry in the context of global challenges.

METHODOLOGY

The methodology of this study was developed based on scientific, theoretical and practical sources. Theoretical, statistical, comparative and expert assessment methods were used to analyze the efficiency of production in food industry enterprises and study ways to improve it. First, through theoretical analysis, the concept of production efficiency, the views of economic schools on efficiency, and the best practices of countries such as Turkey, Germany and the Netherlands were studied. Second, based on statistical analysis, data from the State Statistics Committee of the Republic of Uzbekistan, FAO, World Bank and OECD were analyzed. Production volume, labor productivity, resource efficiency and export-import dynamics for the period 2015–2024 were assessed using indexing and comparison methods. Third, through comparative analysis, the efficiency of the Uzbek food industry was compared with foreign experience, including the Turkish experience in the halal certification system, innovative technologies and state support policy. Fourth, through the expert assessment method, the opinions of specialists, manufacturers, and economists were analyzed and the main factors affecting efficiency were identified.

RESULTS AND DISCUSSION

Current status of the food industry of Uzbekistan. According to the State Statistics Committee (SSC), in 2020–2024, the production volume of the food industry of Uzbekistan showed a steady growth trend. In 2020, the production volume amounted to 39.2 trillion

1 FAO. (2022). The future of food and agriculture: Drivers and triggers for transformation. URL: <https://openknowledge.fao.org/server/api/core/bitstreams/e51e0cf0-4ece-428c-8227-ff6c51b06b16/content>

2 State Committee of the Republic of Uzbekistan on Statistics. (2024). Official report on population of Uzbekistan. URL: Stat.uz

3 World Bank. (2022). Uzbekistan Economic Update: Navigating Global Challenges. URL: <https://www.worldbank.org/en/country/uzbekistan/publication/cem-2025>

soums, while by 2024 this figure reached 56.8 trillion soums. The growth rate averaged 7.8% per year⁴. At the same time, high product costs and low resource efficiency negatively affect economic efficiency.

Labor productivity. Labor productivity in the Uzbek food industry in 2024 was equal to 14,300 US dollars per worker. For comparison, in Turkey this figure was 28,500 US dollars, and in Germany it was 47,200 US dollars⁵. This difference indicates an insufficient level of technological modernization and automation of production processes in our country.

Resource efficiency. There are also significant differences in energy efficiency. According to OECD data, the food industry in the European Union consumes an average of 42 MWh of energy per 1 million euros of production, while in Uzbekistan this figure is 76 MWh⁶. This indicator indicates the wastefulness of resource use, as well as the lack of widespread implementation of energy-saving technologies.

International experience and opportunities for Uzbekistan. The experience of foreign countries determines important directions for Uzbekistan. For example, as a result of the introduction of the Halal certification system in Turkey, not only consumer confidence increased, but also export potential significantly expanded. In 2022, Turkey's poultry production reached 2.2 million tons, making the country one of the world's leading exporters⁷. In the Netherlands, the "Circular Food Economy" model is being successfully implemented. According to it, up to 70 percent of food waste is recycled and returned to production. This approach simultaneously ensures environmental sustainability and significantly reduces costs⁸.

Key challenges for Uzbekistan. The main factors that negatively affect efficiency in the Uzbek food industry are:

1. Technological equipment obsolescence.
2. Low labor productivity.
3. Wasteful use of energy and water resources.

4. Low value added in export products.

Innovative solutions. The following innovative ways are important for the Uzbek food industry to increase efficiency:

- Automation and digitalization: introduction of IoT and artificial intelligence-based control systems into production lines.
- Halal and international certification: expanding export opportunities by adapting the Turkish model.
- Energy efficiency technologies: use of renewable energy sources and establishing an energy audit system in production processes.
- Support for research and innovation: development of food technologies based on public-private partnerships.

The above analysis shows that the Uzbek food industry has great potential for sustainable development in the face of global challenges. However, in order to fully realize this potential, technological modernization, rational use of resources. It is necessary to widely introduce and use innovative approaches.

CONCLUSIONS

The results of the study show that the Uzbek food industry has been showing steady growth in recent years. Although production volumes almost doubled in 2020–2024, the level of efficiency remains significantly lower than that of developed countries. This is due to a number of factors, including technological obsolescence of production processes, low labor productivity, wasteful use of resources, and incomplete compliance with international quality and safety standards⁹.

Analysis of foreign experience shows that Turkey has achieved significant expansion of export opportunities through the widespread introduction of the halal certification system in the food industry¹⁰. The experience of the Netherlands also shows that it has managed to ensure economic and environmental efficiency through waste recycling and efficient use of resources.

4 State Committee of the Republic of Uzbekistan on Statistics. (2025, June 20). Industrial production press release (2020–2024) [Press release]. stat.uz. URL: https://stat.uz/img/press-relizlar/press-reliz-sanoat-20_06_2025-j_-english_p22356.pdf

5 World Bank. (2022). Turkey Economic Monitor: Productivity, Innovation, and Green Transition. URL: <https://openknowledge.worldbank.org/entities/publication/6bfe3775-f21a-5926-89d9-05fd5e4b3287>

6 OECD. (2021). Energy efficiency in food and drink manufacturing. Paris: OECD Publishing. URL: https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/12/energy-efficiency-2021_eb264079/cfb80e1e-en.pdf

7 FAO. (2023). Food and Agriculture Statistics Yearbook 2023. Rome: Food and Agriculture Organization of the United Nations. URL: <https://www.fao.org/statistics/yearbook>

8 Dutch Ministry of Agriculture, Nature and Food Quality. (2022). Circular Agriculture in the Netherlands: Vision and Strategy. The Hague: Government of the Netherlands. URL: <https://www.government.nl/topics/circular-economy/circular-dutch-economy-by-2050>

9 World Bank. (2022). Uzbekistan Country Economic Memorandum: Structural transformation for inclusive growth. Washington, DC: World Bank Group. URL: <https://openknowledge.worldbank.org/entities/publication/2c3adf3e-f349-46a3-b2ea-c44165d9201b>

10 World Bank. (2022). Turkey Economic Monitor: Productivity, Innovation, and Green Transition. Washington, DC: World Bank Group. URL: <https://openknowledge.worldbank.org/handle/10986/38039>

These approaches can also be applied in the conditions of Uzbekistan¹¹. On this basis, the following priority proposals are put forward to increase efficiency in the country's food industry:

1. Technological modernization - introduction of automated production lines and development of management mechanisms based on digital technologies (IoT, ERP systems).

2. Energy and resource efficiency - widespread implementation of energy-saving technologies, use of renewable energy sources and formation of waste processing clusters in the meat and dairy industry.

3. Halal and international standards - strengthening the halal certification system at the institutional level, as well as increasing export potential through the widespread introduction of ISO and HACCP standards.

4. Personnel and scientific research - training qualified specialists in innovative technologies, strengthening cooperation between research institutions and the private sector.

5. State support - encouraging manufacturers through tax incentives, subsidies and soft loans, establishing grants and special programs for enterprises implementing innovative technologies.

The general conclusion is that the Uzbek food industry has great potential for sustainable development in the face of global challenges. To fully realize this potential, technological modernization, rational use of resources, and the widespread introduction of innovative approaches are necessary. In particular, the areas of automation of production, certification based on international standards, energy efficiency, and strengthening of scientific and research activities can bring the

national food industry to a qualitatively new level.

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2. FAO. (2022). The future of food and agriculture: Drivers and triggers for transformation. URL: <https://openknowledge.fao.org/server/api/core/bitstreams/e51e0cf0-4ece-428c-8227-ff6c51b06b16/content>

3. State Committee of the Republic of Uzbekistan on Statistics. (2025, June 20). *Industrial production press release (2020–2024) [Press release]*. stat.uz. URL:https://stat.uz/img/press-relizlar/press-reliz-sanoat-20_06_2025-j_-english_p22356.pdf

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5. World Bank. (2022). Turkey Economic Monitor: Productivity, Innovation, and Green Transition. URL: <https://openknowledge.worldbank.org/entities/publication/6bfe3775-f21a-5926-89d9-05fd5e4b3287>

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11 Dutch Ministry of Agriculture, Nature and Food Quality. (2022). Circular Agriculture in the Netherlands: Vision and Strategy. The Hague: Government of the Netherlands. URL:<https://www.government.nl/topics/circular-economy/circular-dutch-economy-by-2050>